

PARTS LIST

Semiconductors:

- IC1 - LM324 quad op-amp
- IC2 - NE555 timer
- D1, D2 - 1N4007 rectifier diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R4 - 10-kilo-ohm
- R2, R5, R6 - 1-kilo-ohm
- R3 - 1-mega-ohm
- VR1, VR2 - 1-mega-ohm potmeter

Capacitors:

- C1, C3 - 0.1 μ F ceramic disk
- C2 - 10 μ F, 25V electrolytic
- C4, C6 - 100 μ F, 25V electrolytic
- C5 - 0.01 μ F ceramic disk

Miscellaneous:

- CON1-CON3 - 2-pin terminal connector
- S1 - On/off switch
- MIC1 - Electret microphone
- RL1 - 6V-12V, 1 C/O relay
- LED1 - 230V electric bell or 100W, 230V AC lamp
- RL1 - 6V-12V DC power supply

gised for a time delay set by potmeter VR2. Time delay can be set from a few seconds to 30 minutes. Energisation of the relay is indicated by the glowing of LED1.

Relay RL1 is de-energised after

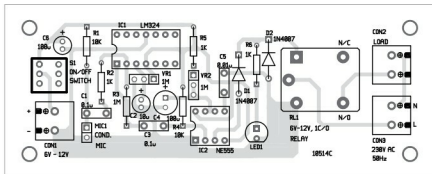


Fig. 3: Component layout of the PCB

the time delay. NE555 can source 200mA current; so, an ordinary relay can be directly connected to IC2's output without an extra transistor driver. Use a good-quality microphone for maximum sensitivity.

Construction and testing

An actual-size, single-side PCB layout for the sound-operated timer is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on the PCB, enclose it in

a suitable plastic box.

The circuit needs a 6V-12V DC power supply. When MIC1 captures the sound made by an intruder, RL1 gets energised. An electric lamp may be connected via N/O contacts of RL1 along with a bell or an alarm. **EFY**



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